



Linux And Bash

Linux & Bash

- Bash is the language that you can write scripts (programs) in linux
- You can create file xyz.sh and then write all commands you need then execute it with
- # bash xyz.sh

Linux & Bash

- First add `#!/bin/bash` in the first line
- Example: `# nano xyz.sh`

```
#!/bin/bash
```

```
ls
```

```
echo "this is red nexus"
```

Then save file as CTRL+X then Y

```
# bash xyz.sh
```

Linux & Bash

To print "Hossam Shady"

```
#!/bin/bash
```

```
name="Hossam Shady"
```

```
echo $name
```

Save then

```
# bash xyz.sh
```

Linux & Bash

To read a file in location /home/xyz.txt

```
#!/bin/bash
```

```
file="/home/xyz.txt"
```

```
cat $file
```

Save and **# bash xyz.sh**

Linux & Bash

- To take argument from user while executing file

```
#!/bin/bash
```

```
echo $1
```

\$1 means the first argument while executing

```
# bash xyz.txt rednexus
```

It will print rednexus

Linux & Bash

- If I need to give two or three arguments while executing

```
#!/bin/bash
```

```
echo "the first name is $1 and second name $2"
```

```
-----
```

```
# bash xyz.sh hossam shady
```

Linux & Bash

- You can make loop in terminal not in a file as
- `# for i in {1..100};do echo "$i" ;done`

It will print all numbers between 1 to 100

If we need to read file lines one by one

```
# for I in `cat xyz.txt`;do echo "$I" ;done
```

Observe (`) is ÷

Linux & Bash

- To get the output of command use `` as
- `# for i in `ifconfig` ;do echo $i ;done`

Linux & regex

- Regex is general identification for character shapes (use regex.io site to help you)
- When you need to specify number you say number without knowing it would be 10 or 9491 or 0
- When you need to catch letter it doesn't matter if it is s or w or a
- As if you need to specify the number of boys and girls , does not matter their names need only the number

Linux & Bash

- Then when you need to catch number then write
- `[0-9]`
- As `[0-9]` is for (digital number)
- `[0-9]{2}` => to catch 2 numbers
- `[0-9][0-9]` => to catch two numbers
- `\s` => to catch space
- `\s{2}` => to catch 2 spaces

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- Suppose that I have text "this is red nexus 101 & new diploma"
- I need to get only numbers then
- `[0-9]{3}`
- I need to get only spaces
- `\s`
- catch any letter or number
- `\w`

Linux & Bash

- When you see[.]Means can be anything (letter , number , special character ?&^%\$#@
- Then what to do to catch real .
- "this is new red teaming \$ new diploma 101 "
- You need to specify everything

.+

- + means any number of characters

Linux & Bash

- "this is new diploma of red . Teaming . hacking"
- To get the .

\\.

For any special character like `$%^&?!@][()/\|`

Add only `\` before it as

`\? \@ \. \[ \) \} \!`

Linux & Bash

- `"https://example.com"`
- `\w{5}\:\/\w{7}\.\w{3}`
- To match only x but in the beginning of any word
- `^x`
- To match x but at the end of any word `x$`

Linux & Bash

```
"this is line one  
This is line two  
This is line three  
This is line four"
```

```
.+e$
```

It will match the line 1 and 3 as they end with (e)

Linux & Bash

- # grep
- To get specific content from file as if you need to get all lines only containing the word "hacking"
- # cat xyz.txt | grep "hacking"
- If you need to get the word hacking but at end of lines
- # cat xyz.txt | grep -E "hacking\$"

Linux & Bash

- Suppose I need to extract all urls containing files .php at the end

"https://google.com/file.txt

https://google.com/file.js

https://google.com/file.php

https://google.com/file.html"

- `# cat urls.txt | grep -E "\.php$"`

Linux & Bash



- Questions